Ten Classical Problems, Run for a Thousand Seconds

A long-duration verification of the ManifoldsX multibody solver

A simulation can hold its constraints and its energy almost perfectly and still end up in the wrong place. To see how often that happens, we ran ten classical problems from the history of mechanics for 1,000 simulated seconds each and compared every run with an exact or independently checked reference. The problems run from Kepler's eccentric orbit to a chain of a thousand masses, and each was picked because it breaks solvers in a different way. ManifoldsX, with its projected HEDOP5 integrator, passes every case, with settings that run at 3 to 83 times real time. The HHT integrator passes several cases outright and others at a smaller step. Three runs that kept their energy to better than 0.05 % ended between 24° and 180° away from the true answer. Those failures are reported next to the successes.

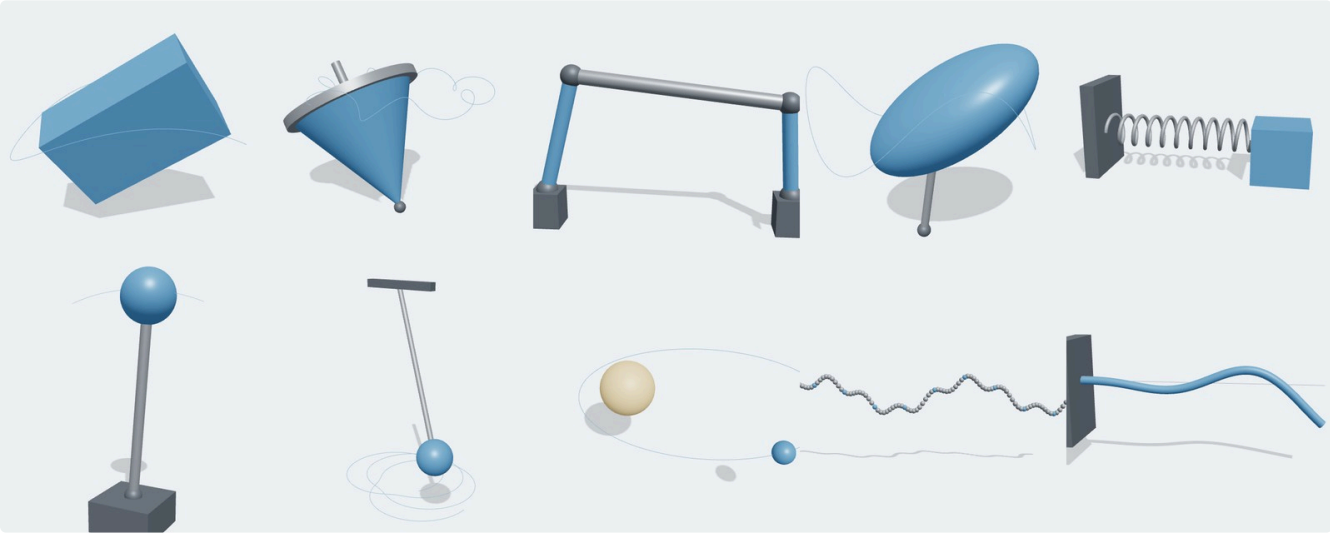


Figure 1. The ten models. Top row: free rigid body, Lagrange top, four-bar linkage, Kovalevskaya top, Duffing slider. Bottom row: near-inverted pendulum, spherical pendulum, Kepler orbit, chain of masses, cantilever beam.

1. Why run for a thousand seconds

Most solver tests are short. A pendulum swings a few times, a top precesses for a few seconds, and the result is compared with a plot or a conservation law. Short tests catch bugs. They say much less about hours of simulated operation, which is what a digital twin, a fatigue study or a long training run actually needs.

Over a long run, errors too small to see in one step pile up in the phase of the motion. Energy and constraint residuals are poor guards against this. A pendulum that arrives a few microseconds late on every swing still has the right energy and still sits exactly on its circle. After a few hundred swings it is simply somewhere else. The only way to catch that is to compare the trajectory itself with one that is known to be right.

Classical mechanics has a short list of problems whose answers are known exactly. Most of them are old, and each comes with a story. We took ten of them and set each one up to press on a particular weakness of numerical integration. Then we ran each for 1,000 s and kept every result, good or bad [1]. The cases appear below roughly in the order they entered the history of mechanics, from Kepler's orbit to the first computer experiment on a chain of masses.

Related work

Most multibody codes integrate the constrained equations of motion with implicit methods that came out of structural dynamics. The Hilber–Hughes–Taylor (HHT) method [2] and the generalized-α family [3] add controllable numerical damping to the Newmark scheme. Negrut and colleagues described an HHT implementation for the index-3 equations of multibody dynamics [4], and Arnold and Brüls proved second-order convergence of generalized-α for constrained systems [5]. Keeping constraints from drifting has its own history, from Baumgarte's stabilization [6] to the projection methods treated by Hairer and Wanner [7]. Geometric integration [8] explains why some methods keep the long-time character of a motion better than others. Celledoni and co-authors showed how the free rigid body can be computed exactly with elliptic functions [9], which is the kind of reference we rely on here. On the engine side, MuJoCo [10] and Chrono [11] made fast, general multibody simulation widely available.

Our question is narrower and more practical. After a long run, how far is the computed trajectory from the exact one? And does a good energy or constraint record predict that distance? We answer it case by case, failures included.

2. How the runs were set up

Every run asks for 1,000 s of simulated time at a fixed step. There is no error control and no step cutting, so a run either finishes at the requested step or stops.

Two integrators were used. HHT runs use $\alpha = 0.1$ on the index-3 equations, without projecting positions or velocities back onto the constraints. HEDOP5 runs use the production explicit path with projection onto the constraints at a tolerance of 10^{-11}. Newton iterations converge to 10^{-9} on accelerations and 10^{-10} on positions.

Each case is scored on the quantity that matters for it, against a pass mark listed in Table 1. A run must also finish, keep position constraints below 10^{-8} and velocity constraints below 10^{-4}, and, unless it is driven, keep its energy within 1 %. The pass marks are loose compared with the errors we hoped to see. A pass means the trajectory is right, not just close.

Table 1. What each case is scored on

Case	Scored quantity	Pass mark
Kepler orbit	Position	10 mm (1 % of the semi-major axis)
Near-inverted pendulum	Direction of the bob	1°
Flexible beam	Transverse displacement at every node	1 % of the summed mode amplitudes
Euler free body	Full orientation	1°
Four-bar near toggle	Drive torque	5 % of its peak
Lagrange top	Full orientation	1°
Spherical pendulum	Direction of the bob	1°
Kovalevskaya top	Full orientation, and drift of its extra invariant	1°, and 1 %
Duffing oscillator	Displacement and phase	1 % of the amplitude, and 1°
Chain of 1,000 masses	Displacement of every mass	1 % of the summed mode amplitudes

3. The ten problems

3.1 Kepler's orbit

In 1609 Johannes Kepler published *Astronomia nova*. In it he argued, from years of Tycho Brahe's observations of Mars, that a planet moves on an ellipse with the Sun at one focus and sweeps out equal areas in equal times [12]. Newton showed in the *Principia* that an inverse-square attraction produces exactly such orbits [13]. Kepler also wrote down the equation that carries his name, $M = E - e \sin E$, which turns a time into a position on the ellipse. It has no closed-form solution, but it can be solved to machine precision in a few iterations. That makes the orbit one of the cleanest references in mechanics.

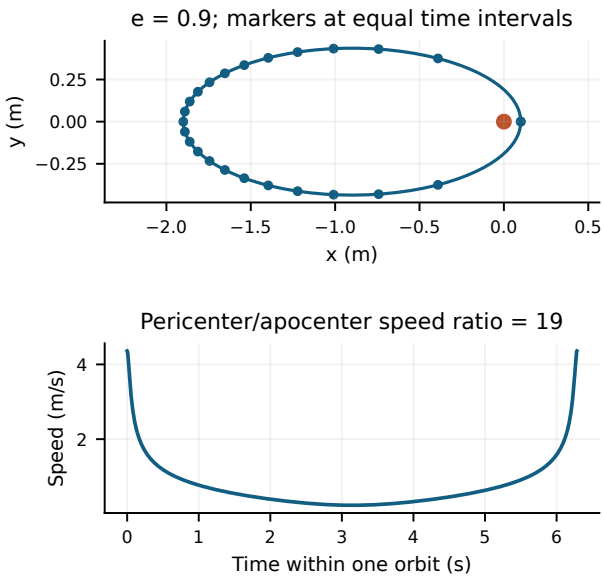


Figure 2. The exact orbit with markers at equal time intervals, and the speed over one period. The markers spread out near the focus, where the body moves fastest.

An eccentric orbit is the textbook example of motion that is slow most of the time and fast for a moment. Comets behave this way, and so do mechanisms in which a link whips through a short part of its cycle. At an eccentricity of 0.9 the body moves 19 times faster at its closest approach than at its farthest. With a fixed step, most of the error is made during that brief pass, and every pass shifts the phase a little. Over about 159 revolutions the shifts add up.

Unit mass, gravitational parameter and semi-major axis; eccentricity 0.9. The body starts at its closest point, (0.1, 0, 0) m, with speed √19 m/s. The applied force is $-\mathbf{r}/|\mathbf{r}|^3$. The period is 2π s. The reference solves Kepler's equation for the eccentric anomaly and gives exact position and velocity.

HHT at 1 ms finished the run but ended 1.33 m from the exact position, more than the semi-major axis itself. An error that size puts the body on a different part of its orbit altogether. HEDOP5 passed at 1 ms with 0.30 mm and at 2 ms with 7.9 mm. At 4 ms it missed the 10 mm mark with 35 mm, although its energy stayed close. The lesson is simple: the pass near the focus, not the average step, decides how fine the step must be.

3.2 The pendulum near the top

Galileo believed that a pendulum takes the same time for a swing whatever its size. Christiaan Huygens showed in *Horologium oscillatorium* (1673) that this is only approximately true on a circular arc, and he designed a cycloidal suspension for his clocks to correct it [14]. The period grows with amplitude. As the release point approaches the upside-down position, the period grows without limit, because a pendulum placed exactly at the top would balance there forever. The motion that just fails to go over the top lies next to the separatrix, the boundary between swinging and rotating.

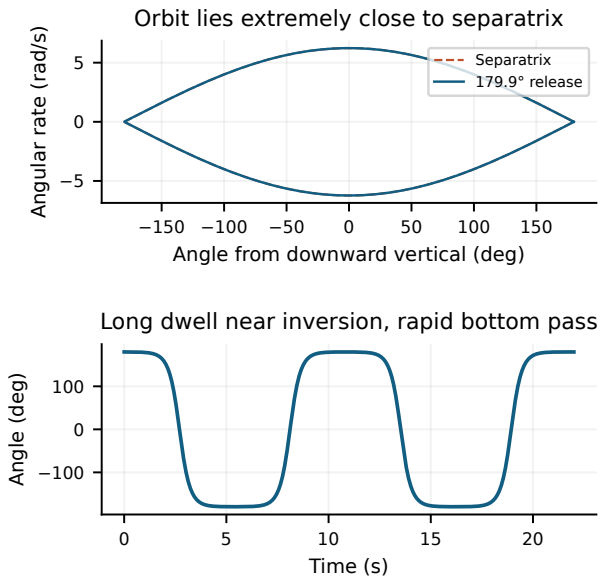


Figure 3. Phase portrait and angle history from the exact solution. The orbit lies so close to the separatrix that the two curves overlap at this scale.

Near the top the pendulum almost stops, and the time it spends there depends very strongly on its energy. A tiny energy error therefore becomes a large timing error. Over-centre linkages, snap-through panels and balancing robots all live near points like this.

Mass 1 kg, pivot to centre of mass 1 m, inertia about the centre 0.01 kg m^2, gravity 9.81 m/s^2. Released from rest just short of upside down, with energy 1.5×10^{-5} J below the separatrix. Each swing takes 10.8 s, more than five times the small-swing period. The reference is $\theta(t) = 2\,\mathrm{asin}[k\,\mathrm{sn}(K - \Omega t \mid k^2)]$, checked at late crossings against 60-digit evaluation of the elliptic functions [15].

This case gave the starkest result of the study. HHT at 1 ms kept the energy to within 5.7×10^{-3} % of its starting value, and at the end the bob pointed the opposite way: an error of 180°. An energy check would have passed it. HEDOP5 passed at 1 ms with a direction error of 2.4×10^{-3} degrees and at 4 ms with 0.74°, still inside the 1° mark.

3.3 A cantilever that bends while it moves

Galileo asked how strong a cantilever is in his *Two New Sciences* of 1638. About a century later Euler and Daniel Bernoulli worked out how such a beam bends and vibrates, and their theory is still the first tool an engineer reaches for. Robot arms, booms and deployable structures are now light enough to bend while they move. A multibody solver that includes flexible members has to get both the stiffness and the inertia of each element right, including the rotary inertia, or the higher modes slowly drift out of step.

A single vibrating mode can look right even when the rotary inertia is wrong. So we excited three modes at once and let them run together.

Eight production Beam6D elements form a 1 m cantilever with a square section 0.1 m wide. Young's modulus is 1,000 Pa, deliberately soft, and density 1,000 kg/m^3. No damping. Initial velocities excite the first three modes with amplitudes 10, 4 and 2 µm; their frequencies are 0.10, 0.60 and 1.6 rad/s. The reference is the exact modal solution of the same discrete model, linearized, with the same lumped translational and rotary masses.

HHT at 4 ms stayed within 1.6×10^{-8} m. HEDOP5 at 16 ms stayed within 9.5×10^{-13} m, at 62 times real time. Halving the amplitude cut that small gap eightfold, to 1.2×10^{-13} m. That is the signature of the geometric nonlinearity in the production beam, which the linear reference leaves out, and not an integration floor. The result shows agreement with the discrete model. It is not a mesh-convergence study of the continuous beam.

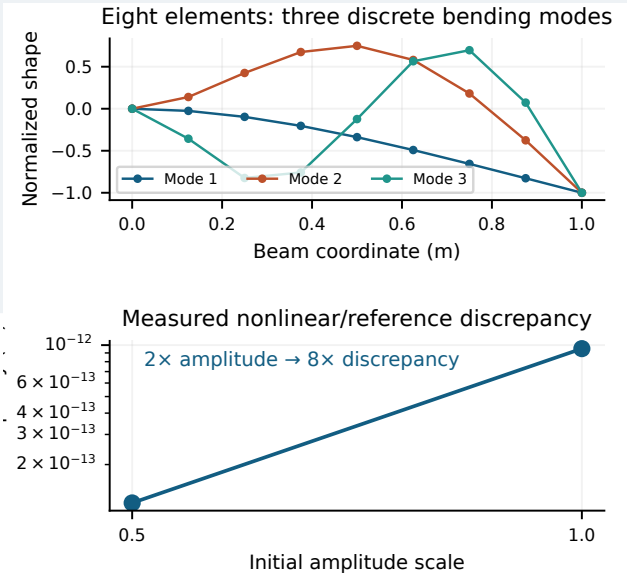


Figure 4. Mode shapes of the discrete beam, and the measured gap between HEDOP5 and the linear reference at two amplitudes. Shapes are normalized; the physical amplitudes are 10, 4 and 2 µm.

3.4 Euler's free body and the flipping wing nut

Euler set out the equations of a rotating rigid body, together with the idea of principal axes, in *Theoria motus corporum solidorum seu rigidorum* of 1765 [16]. A consequence every student meets later is that spin about the largest or smallest axis is stable and spin about the middle axis is not. In 1985, on the mission that brought the dead Salyut 7 station back to life, cosmonaut Vladimir Dzhanibekov spun a wing nut off its bolt and watched it flip over every few seconds with nothing touching it. The effect is often named after him.

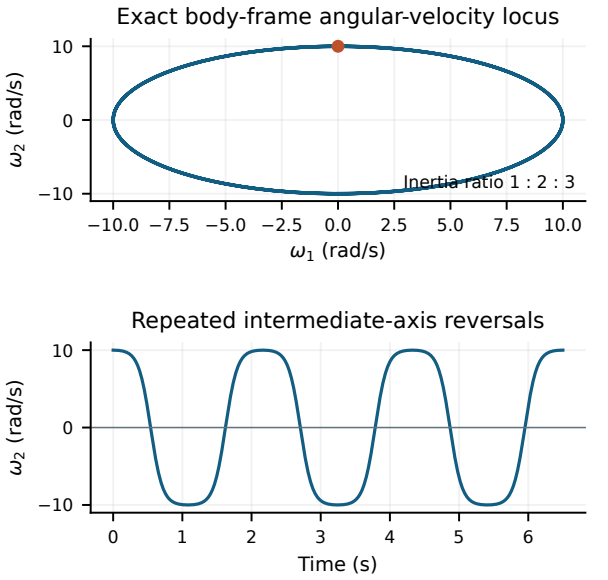


Figure 5. Exact solution: the angular-velocity path in the body frame, and the repeated sign changes of the middle-axis component. Each sign change is one flip.

Satellites, tumbling debris, thrown tools and drones in free flight all rotate this way. The case has no contact, no joints and no forcing, so the only thing that can go wrong is the rotation update itself. Each flip passes close to a separatrix, so a small error moves the time of the next flip. After hundreds of flips, a tumble that looks entirely plausible can be upside down.

Principal inertias 1, 2 and 3 kg m^2; initial body angular velocity (0, 10, 1) rad/s. The angular velocity repeats every 2.16 s, so 1,000 s contains about 462 periods and 924 flips. The reference uses Jacobi elliptic functions for the angular velocity and elliptic integrals of the third kind for the attitude [9].

HHT at 1 ms kept the energy to within 2.8×10^{-3} % and ended 125° away from the true orientation. At 0.25 ms it passed with 0.63°, which shows the error is controlled by the step. HEDOP5 at 4 ms stayed within 0.022° through all 924 flips, at 69 times real time.

3.5 The four-bar linkage at the edge of toggle

In 1784 James Watt patented a four-bar linkage that guided the piston rod of his engine along a nearly straight line, and he soon refined it into his “parallel motion”. In a letter to his son years later he wrote that he was prouder of the parallel motion than of any other mechanical invention he had made. A century after Watt, Franz Grashof gave the rule that tells, from the four lengths alone, whether a link can turn all the way round [17]. If the shortest and longest links together are shorter than the other two, one link can rotate fully. If the two sums are equal, once per turn the links fall into a single straight line. That is the toggle, or dead-centre, position.

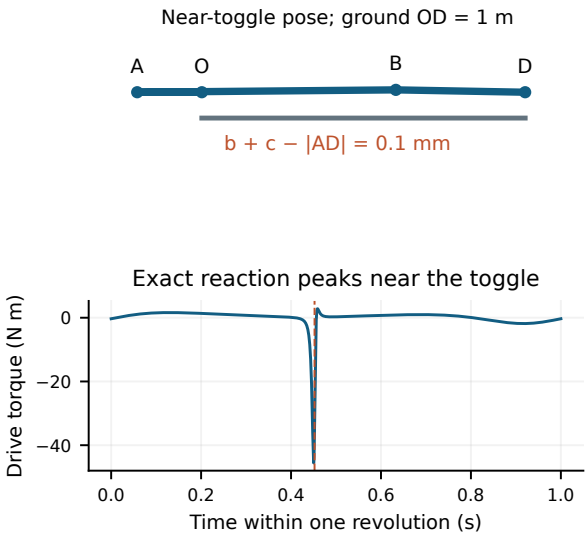


Figure 6. The linkage drawn to scale at its closest approach to toggle, where the 0.1 mm gap is too small to see, and the exact drive torque over one revolution.

Toggle is useful. Near it a large force can be held with a small one, which is why clamps, presses and crushers are built around it. It is also where solvers get into trouble. At toggle the constraint equations lose a direction, and just before it they are badly conditioned. A position error too small to see on screen becomes a large error in acceleration and in the joint forces. That is why this case is scored on the drive torque, not on the motion.

Ground, crank, coupler and rocker are 1, 0.2, 0.8 and 0.4 m long, so the two Grashof sums are equal to within a tenth of a millimetre. Moving masses are 1, 2 and 1.5 kg. The crank is driven at one revolution per second, starting at 0.3 rad, with gravity off. Once per revolution the linkage comes within 0.1 mm of lying flat in one line, and the run passes there 1,000 times. The reference takes the positive branch of the circle intersection, differentiates the loop closure twice, and finds the drive torque from the rate of change of kinetic energy.

Both integrators pass at 1 ms. HHT stays within 1.8 % of the peak torque and HEDOP5 within 3.6×10^{-3} %. The failures along the way were more instructive. With position projection set very tight, at 10^{-14}, the HEDOP5 run stopped at 10.4 s. At that moment the spacing between neighbouring double-precision numbers near the crank angle was 1.4×10^{-14} rad, so the projection was asking for more accuracy than the number format can hold. Relaxing it to 10^{-11} fixed the problem; relaxing only the velocity projection did not. A 4 ms run rejected a step at 3.45 s, and we did not diagnose that one separately. The passing run is the slowest in the study, at 3.1 times real time.

3.6 Lagrange’s heavy top

Lagrange solved the symmetric heavy top, a spinning top with its centre of mass on its axis, in the *Mécanique analytique* of 1788 [18]. It is the second of the three classical tops that can be solved exactly, and the most familiar: anyone who has played with a toy top has watched it lean, circle and nod. The same physics runs gyroscopes, gimbals and tools spinning on a pivot.

Many solver tests use the gentle case of steady precession, where the tilt never changes. We chose a hard one. Our top sweeps from 69° to 166° from the vertical every 0.715 s, so gravity torque, pivot reaction and fast spin all change together, about 1,400 times over the run.

Mass 1 kg, centre of mass 0.2 m from the pivot, inertia diag(0.02, 0.02, 0.01) kg m^2. Initial tilt 1.2 rad and axial spin 10 rad/s, with no initial precession or nutation. The reference follows the tilt from a cubic first integral in cos θ and rebuilds precession and spin with elliptic integrals [19].

HHT at 1 ms held energy to 0.045 % and missed the orientation by 24.5°. At 0.25 ms it passed with 0.61°. HEDOP5 at 4 ms stayed within 0.020°, at 56 times real time.

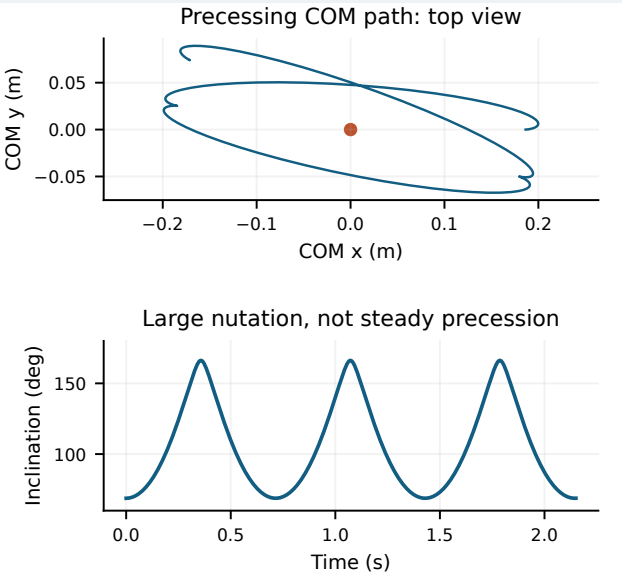


Figure 7. Exact solution: path of the centre of mass seen from above, and the inclination over about three nutation periods. The pivot is at the origin.

3.7 The spherical pendulum

In 1851 Léon Foucault hung a heavy bob from the dome of the Panthéon in Paris and let the public watch the Earth turn underneath it. His pendulum was spherical: the bob was free to move anywhere on a sphere, not only in a plane. A load on a crane hook or under a drone moves the same way. A planar pendulum cannot reveal errors in how the sideways and up-and-down motions couple. A spherical one can, because its path turns a little with every swing.

The body of Section 3.2 on a spherical joint, started 0.7 rad from the upward vertical with an azimuth rate of 0.6 rad/s. The reference uses a cubic first integral in cos θ, Jacobi functions for the inclination and elliptic integrals of the third kind for the azimuth, with the cubic roots computed to 60 digits. Five long-time checks agree within 3.3×10^{-12} m.

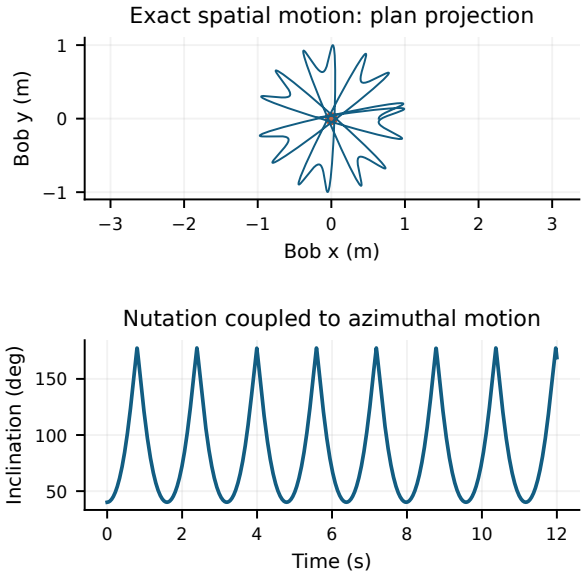


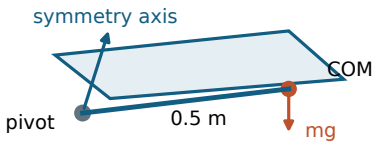
Figure 8. Exact solution: the bob path seen from above, and its inclination from the upward vertical. The path turns a little with every swing.

HHT at 1 ms missed the mark with 3.3° and passed at 0.5 ms with 0.24°. HEDOP5 at 4 ms stayed within 4.1×10^{-5} degrees, at 59 times real time.

3.8 Kovalevskaya's top

After Euler and Lagrange, the next solvable heavy top took a century to find. In 1888 the Paris Academy awarded its Prix Bordin to Sofia Kovalevskaya, and raised the prize from 3,000 to 5,000 francs, for a top nobody had thought to look at [20]. Two of its principal moments are equal and twice the third, and its centre of mass sits in the equatorial plane. Her solution needed hyperelliptic functions. No further general solvable case has been found since.

Numerically it is the hardest of the classical tops. Gravity acts off the axis, and all three components of the angular velocity interact. It also offers a sharp check. Besides energy and angular momentum, the motion keeps a fourth-degree quantity K constant, and keeping the energy right does not keep K right.



Equatorial COM + 2:2:1 pivot inertia

Coupled rotation about all three axes

$K = (p^2 - q^2 - \gamma_x)^2 + (2pq - \gamma_y)^2$

An extra exact invariant tests the coupled dynamics

Unit mass; inertia about the pivot diag(2, 2, 1) kg m^2; gravity 2 m/s^2; centre of mass at (0.5, 0, 0) m. Initial orientation $R_y(0.7)R_z(0.4)$ and body angular velocity (0.6, 1.1, 0.8) rad/s. With $\omega = (p, q, r)$ and γ the vertical in body axes, $K = (p^2 - q^2 - \gamma_x)^2 + (2pq - \gamma_y)^2$. We did not implement the hyperelliptic solution. The trajectory reference comes from independent, refined Euler–Poisson integrations that agree with each other within 1.9×10^{-9} degrees. The invariants themselves are exact.

Figure 9. Geometry of the top (schematic) and its extra invariant. The centre of mass lies in the equatorial plane, not on the axis as in the Lagrange top.

All three settings pass. HHT at 1 ms was within 0.29°. HEDOP5 at 4 ms was within 8.7×10^{-8} degrees, with a relative drift in K of 3.0×10^{-13}. At 83 times real time it was also the fastest run in the study.

3.9 Duffing's stiffening spring

Georg Duffing was an engineer and inventor, not a professional mathematician. In 1918 he published a 134-page booklet on forced vibration with a frequency that changes with amplitude, written to explain odd resonances he had seen in pendulum-like machines [21]. His equation with a cubic spring became the standard model of nonlinear vibration, and from the 1970s a favourite example in chaos theory [22].

Rubber bushings, elastic end stops and stiffening structures all behave like his spring. The stiffness changes with displacement, so the frequency changes with amplitude, and any numerical damping turns into a phase error.

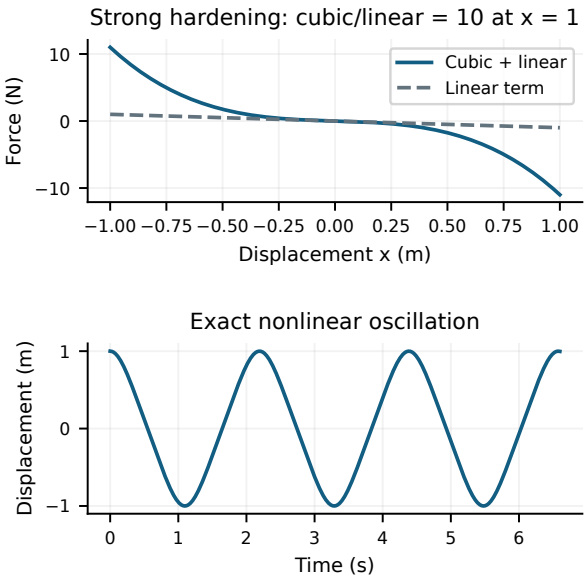


Figure 10. The cubic restoring force against its linear part, and the exact displacement.

A unit-mass slider on a translational joint obeys $\ddot{x} + x + 10x^3 = 0$, released from rest at $x = 1$ m, where the tangent stiffness is 31 times its linear value. The exact solution is $x(t) = \mathrm{cn}(\sqrt{11}\,t \mid 5/11)$. The period is 2.19 s, so the run covers 456 cycles.

All three settings pass. HHT at 1 ms was within 3.5 mm, with a phase lag of 0.24° at the last crossing. HEDOP5 at 4 ms was within 0.83 µm, with a lag of 5.6×10^{-5} degrees, at 38 times real time. At the same 1 ms step HHT is the faster method, 35 against 16 times real time, while HEDOP5 is far more accurate. Cost and accuracy have to be judged together.

3.10 A chain of a thousand masses

In 1955 Enrico Fermi, John Pasta and Stanislaw Ulam, with the programming done by Mary Tsingou, ran one of the first numerical experiments in physics on the MANIAC computer at Los Alamos [23], [24]. Their model was a line of 16 to 64 masses joined by slightly nonlinear springs. They expected the energy put into one vibration mode to spread evenly over all the others. Instead it came back, almost entirely, to the mode where it started. The surprise helped launch nonlinear science, and the habit of using computers to do experiments.

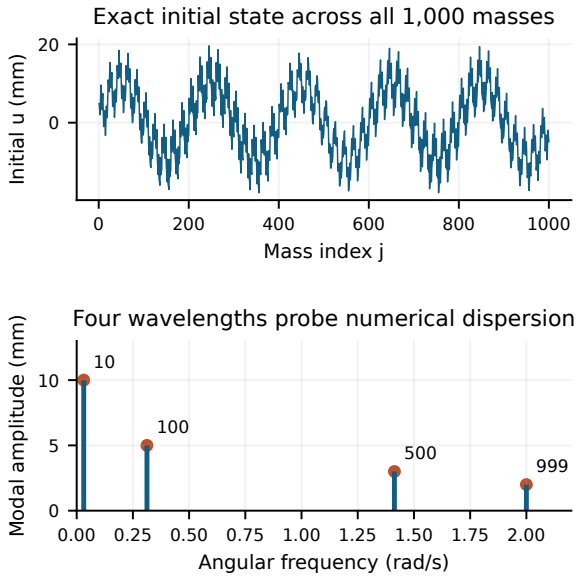


Figure 11. Initial displacement of the chain and the frequencies of the four excited modes. All 1,000 masses are scored.

Our chain is much longer, and its springs are linear on purpose. The exact motion is then a sum of independent modes, so any energy that moves from one mode to another, or any mode that drifts out of phase, is a numerical error. Large models such as long cables and finite element meshes have many modes at very different frequencies, and the fastest ones are where a fixed step loses accuracy first.

One thousand unit masses joined by 1,001 springs of stiffness 1 N/m between fixed anchors; 6,000 velocity degrees of freedom in the production model. Modes 10, 100, 500 and 999 are excited with amplitudes 10, 5, 3 and 2 mm; mode 999 is close to the highest frequency the chain can carry. Starting from rest, the exact motion is $u_j(t) = \Sigma_p A_p \sin(\pi j p/1001) \cos(\omega_p t)$, with $\omega_p = 2 \sin(\pi p/2002)$. Every mass is scored at every saved time.

HHT at 20 ms was fast but ended 0.99 mm out, above the 0.2 mm mark. HEDOP5 at 80 ms stayed within 0.14 µm at 4.3 times real time, and at 20 ms within 1.4×10^{-10} m at about real time. The test is longitudinal only. It does not cover contact or long chains of hinges.

4. Results at a glance

Table 2 lists, for every case, the passing HEDOP5 setting reported above and the best HHT result we recorded. Figure 12 shows the speed of the HEDOP5 runs.

Table 2. Largest error over 1,000 s. Speed is simulated time divided by wall-clock time.

Case	HEDOP5 step	HEDOP5 error	Speed	Best HHT result
Kepler orbit	1 ms	0.30 mm	30×	1.33 m at 1 ms (fail)
Near-inverted pendulum	1 ms	2.4×10^{-3} deg	17×	180° at 1 ms (fail)
Flexible beam	16 ms	9.5×10^{-13} m	62×	1.6×10^{-8} m at 4 ms
Euler free body	4 ms	0.022°	69×	0.63° at 0.25 ms
Four-bar near toggle	1 ms	3.6×10^{-3} % of peak torque	3.1×	1.8 % at 1 ms
Lagrange top	4 ms	0.020°	56×	0.61° at 0.25 ms
Spherical pendulum	4 ms	4.1×10^{-5} deg	59×	0.24° at 0.5 ms
Kovalevskaya top	4 ms	8.7×10^{-8} deg	83×	0.29° at 1 ms
Duffing oscillator	4 ms	0.83 µm	38×	3.5 mm at 1 ms
Chain of 1,000 masses	80 ms	0.14 µm, every mass	4.3×	0.99 mm at 20 ms (fail)

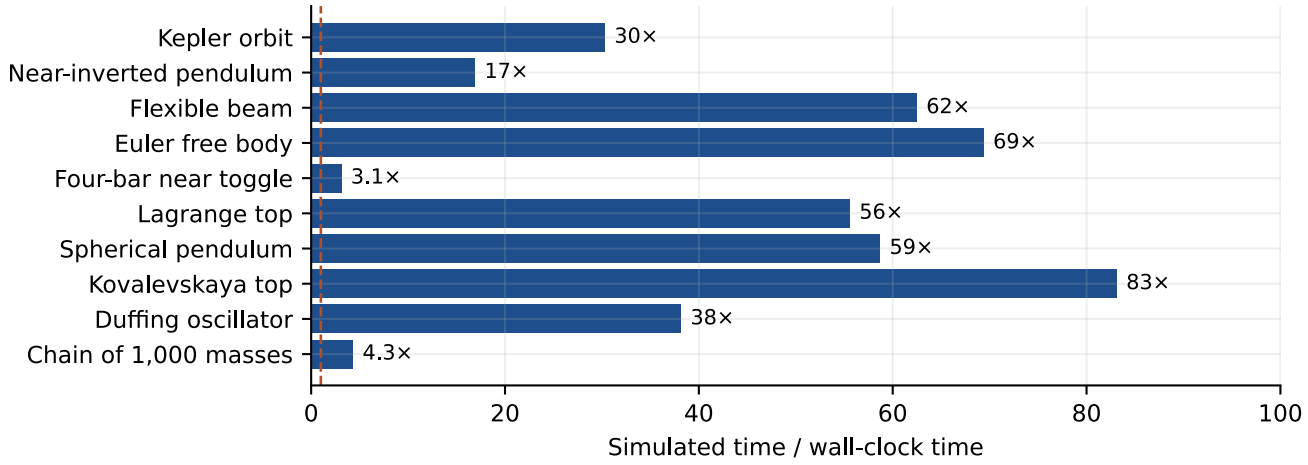


Figure 12. Speed of the HEDOP5 settings in Table 2. The dashed line marks real time. The figures include start-up, per-step diagnostics and file output.

5. What the long runs taught us

Energy is not accuracy

Three runs make the point better than any argument. The HHT free body at 1 ms kept its energy to 2.8×10^{-3} % and ended 125° off. The Lagrange top at the same step kept it to 0.045 % and ended 24.5° off. The near-inverted pendulum kept it to 5.7×10^{-3} % and ended pointing the wrong way. All three would have passed a check on energy and constraints. The first two pass the trajectory check at a smaller step. For the pendulum we did not record a passing HHT step.

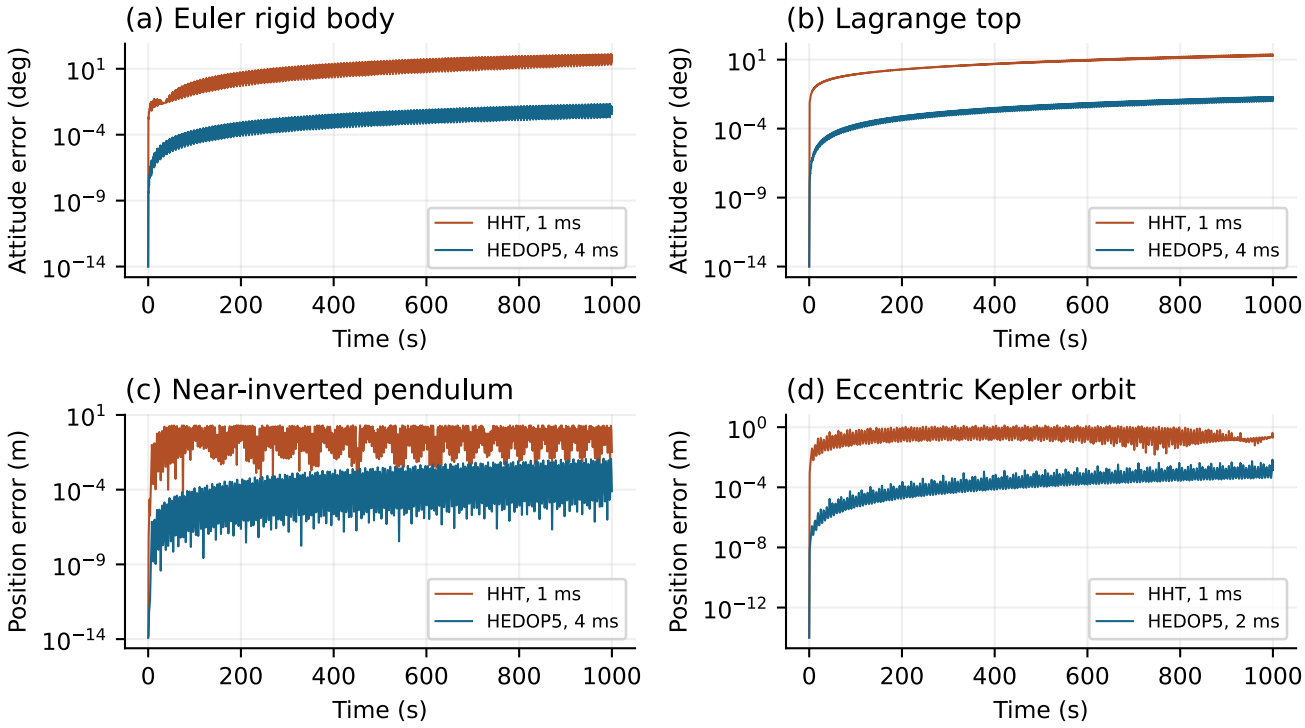


Figure 13. Error over time for four cases, HHT at 1 ms against HEDOP5, with no phase alignment or resynchronization. The curves use retained samples; the maxima reported in the text use every scored sample.

A pass belongs to a setting, not to a method

When we say every case passes, we mean each one has at least one recorded setting that meets its mark. We do not mean every integrator passes at every step. In the five cases run last we completed 18 settings, including step refinements and amplitude checks. Thirteen passed and five failed. The failures were HHT on the near-inverted pendulum, the spherical pendulum, the Kepler orbit and the chain at their baseline steps, and HEDOP5 on the Kepler orbit at 4 ms. Two of them were fixed by a smaller step: spherical HHT at 0.5 ms and Kepler HEDOP5 at 2 ms.

Precision has a floor

The four-bar run that stopped at 10.4 s did not fail for lack of physics. It failed because we asked the projection for a tolerance finer than double precision allows near a large angle. The fix was a sensible tolerance, not a smaller step.

Cost and accuracy belong together

The cheapest step is not the cheapest answer. In the Duffing case HHT runs faster per simulated second at 1 ms, but HEDOP5 at 4 ms is both faster and about four thousand times more accurate. Choosing a method on step cost alone would have picked the wrong one.

6. How the references were checked

A benchmark is only as good as its reference. Eight cases have exact solutions written with elliptic functions, Kepler’s equation or sums of modes. The Kovalevskaya top uses exact invariants and a refined numerical trajectory. The beam uses the exact modal solution of its own linearized discrete model.

Each reference was checked by a different route: short comparisons with an ODE solver, quadrature checks of the elliptic integrals, derivatives of attitude and centre of mass, power balance for the four-bar, eigenvalue relations for the chain and beam, and 60-digit evaluation of special functions for the pendulums. For Duffing, the exact solution satisfies its equation to 3.6×10^{-15}, and an independent energy calculation agrees with the engine to 6.7×10^{-15} J. The near-inverted pendulum reference agrees with its high-precision check to 1.5×10^{-9} m over 104 samples. To test the tester, we gave the scorer a deliberately truncated trajectory, and it rejected it.

Constraint and energy maxima are taken at every step. Trajectory errors are taken at saved times: every step for the four-bar, and every 4 to 20 ms in the other cases.

7. Limits and reproducibility

Every number here describes one model, one setting and one machine. The study does not test contact, friction or long chains of hinges, and it does not establish a scaling law. Speed includes start-up, per-step diagnostics and file output, and excludes building the reference and scoring. Runs were timed one at a time after compilation, but some early HHT refinements overlapped with other jobs, so speed ratios against those runs are not controlled. Results from the earlier stages were not re-measured after the cubic and inverse-square spring forces were added.

The later runs were built on Linux x86-64 with GCC 14.2.0, C++17, −O2 and −DNDEBUG. Scoring used NumPy 2.3.5, SciPy 1.17.0 and pandas 2.2.3. Model construction, reference code, scoring scripts, compact error histories and build records are available on request [1]. The scripts rebuild every model and regenerate every trajectory rather than relying on stored output.

8. Closing

These ten problems span four centuries, and none of them is hard to set up. What makes them useful is that each has a known answer and each finds a different weak spot. Run long enough, they separate a solver that looks right from one that is right. ManifoldsX passes every case with its projected HEDOP5 path, mostly by a wide margin. The failures we recorded point to specific causes: the step near a fast pass, phase near a separatrix, and the limit of floating-point projection near a toggle. We will keep adding cases as we find problems with honest references.

References

1. ManifoldsX long-duration benchmark records: model construction, reference implementations, scoring scripts and run provenance. Available on request.
2. H. M. Hilber, T. J. R. Hughes and R. L. Taylor, “Improved numerical dissipation for time integration algorithms in structural dynamics,” *Earthquake Engineering and Structural Dynamics*, vol. 5, pp. 283–292, 1977.
3. J. Chung and G. M. Hulbert, “A time integration algorithm for structural dynamics with improved numerical dissipation: the generalized-α method,” *Journal of Applied Mechanics*, vol. 60, no. 2, pp. 371–375, 1993.
4. D. Negrut, R. Rampalli, G. Ottarsson and A. Sajdak, “On an implementation of the Hilber–Hughes–Taylor method in the context of index 3 differential-algebraic equations of multibody dynamics,” *Journal of Computational and Nonlinear Dynamics*, vol. 2, no. 1, pp. 73–85, 2007.
5. M. Arnold and O. Brüls, “Convergence of the generalized-α scheme for constrained mechanical systems,” *Multibody System Dynamics*, vol. 18, no. 2, pp. 185–202, 2007.
6. J. Baumgarte, “Stabilization of constraints and integrals of motion in dynamical systems,” *Computer Methods in Applied Mechanics and Engineering*, vol. 1, no. 1, pp. 1–16, 1972.

7. E. Hairer and G. Wanner, *Solving Ordinary Differential Equations II: Stiff and Differential-Algebraic Problems*, 2nd ed. Springer, 1996.
8. E. Hairer, C. Lubich and G. Wanner, *Geometric Numerical Integration*, 2nd ed. Springer, 2006.
9. E. Celledoni, F. Fassò, N. Säfström and A. Zanna, "The exact computation of the free rigid body motion and its use in splitting methods," *SIAM Journal on Scientific Computing*, vol. 30, no. 4, pp. 2084–2112, 2008.
10. E. Todorov, T. Erez and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2012, pp. 5026–5033.
11. A. Tasora, R. Serban, H. Mazhar, A. Pazouki, D. Melanz, J. Fleischmann, M. Taylor, H. Sugiyama and D. Negrut, "Chrono: An open source multi-physics dynamics engine," in *High Performance Computing in Science and Engineering*, Lecture Notes in Computer Science. Springer, 2016, pp. 19–49.
12. J. Kepler, *Astronomia nova*. Heidelberg, 1609.
13. I. Newton, *Philosophiæ naturalis principia mathematica*. London, 1687.
14. C. Huygens, *Horologium oscillatorium*. Paris, 1673.
15. NIST Digital Library of Mathematical Functions, Chapter 22, "Jacobian elliptic functions." dlmf.nist.gov/22.
16. L. Euler, *Theoria motus corporum solidorum seu rigidorum*. Rostock and Greifswald, 1765.
17. F. Grashof, *Theoretische Maschinenlehre*, vol. 2. Leipzig, 1883.
18. J.-L. Lagrange, *Mécanique analytique*. Paris, 1788.
19. H. Goldstein, C. Poole and J. Safko, *Classical Mechanics*, 3rd ed. Addison-Wesley, 2002.
20. S. Kowalevski, "Sur le problème de la rotation d'un corps solide autour d'un point fixe," *Acta Mathematica*, vol. 12, pp. 177–232, 1889.
21. G. Duffing, *Erzwungene Schwingungen bei veränderlicher Eigenfrequenz und ihre technische Bedeutung*. Braunschweig: Vieweg, 1918.
22. I. Kovacic and M. J. Brennan, eds., *The Duffing Equation: Nonlinear Oscillators and their Behaviour*. Wiley, 2011.
23. E. Fermi, J. Pasta and S. Ulam, "Studies of nonlinear problems," Los Alamos Scientific Laboratory report LA-1940, 1955.
24. T. Dauxois, "Fermi, Pasta, Ulam, and a mysterious lady," *Physics Today*, vol. 61, no. 1, pp. 55–57, 2008.